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Diagnostic accuracy of quantitative flow ratio for assessment of coronary stenosis significance from a single angiographic view: A novel method based on bifurcation fractal law

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Abstract

Objectives: We aimed to evaluate the diagnostic accuracy of computation of fractional flow reserve (FFR) from a single angiographic view in patients with intermediate coronary stenosis.

Background: Computation of quantitative flow ratio (QFR) from a single angiographic view might increase the feasibility of routine use of computational FFR. In addition, current QFR solutions assume a linear tapering of the reference vessel size, which might decrease the diagnostic accuracy in the presence of the physiologically significant bifurcation lesions.

Methods: An artificial intelligence algorithm was proposed for automatic delineation of lumen contours of major epicardial coronary arteries including their side branches. A step-down reference diameter function was reconstructed based on the Murray bifurcation fractal law and used for QFR computation. Validation of this Murray law-based QFR (μ QFR) was performed on the FAVOR II China study population. The μ QFR was computed separately in two angiographic projections, starting with the one with optimal angiographic image quality. Hemodynamically significant coronary stenosis was defined by pressure wire-derived FFR ≤ 0.80 .

Results: The μ QFR was successfully computed in all 330 vessels of 306 patients. There was excellent correlation ($r = 0.90$, $p < .001$) and agreement (mean difference = 0.00 ± 0.05 , $p = .378$) between μ QFR and FFR. The vessel-level diagnostic accuracy for μ QFR to identify hemodynamically significant stenosis was 93.0% (95% CI: 90.3 to 95.8%), with sensitivity, specificity, positive predictive value, negative predictive value, positive likelihood ratio, and negative likelihood ratio of 87.5% (95% CI: 80.2 to 92.8%), 96.2% (95% CI: 92.6 to 98.3%), 92.9% (95% CI: 86.5 to 96.9%), 93.1% (95% CI: 88.9 to 96.1%), 23.0 (95% CI: 11.6 to 45.5), 0.13 (95% CI: 0.08 to 0.20), respectively. Use of suboptimal angiographic image view slightly decreased the diagnostic accuracy of μ QFR (AUC = 0.97 versus 0.92, difference = 0.05, $p < .001$). Intra- and inter-observer variability for μ QFR computation was 0.00 ± 0.03 , and 0.00 ± 0.03 , respectively. Average analysis time for μ QFR was 67 ± 22 s.

Abbreviations: AI, artificial intelligence; AUC, area under the ROC curve; DS%, percent diameter stenosis; FFR, fractional flow reserve; ICC, intraclass correlation coefficient; QCA, quantitative coronary angiography; QFR, quantitative flow ratio; ROC, receiver operating characteristic; μ QFR, Murray law-based quantitative flow ratio.

Conclusions: Computation of μ QFR from a single angiographic view has high feasibility and excellent diagnostic accuracy in identifying hemodynamically significant coronary stenosis. The short analysis time and good reproducibility of μ QFR bear potential of wider adoption of physiological assessment in the catheterization laboratory.

KEYWORDS

artificial intelligence, coronary angiography, fractional flow reserve, quantitative flow ratio

1 | INTRODUCTION

Fractional flow reserve (FFR) is the current gold standard for evaluating the physiological significance of non-occlusive coronary stenosis.¹ Nevertheless, the clinical adoption of FFR is limited² due to multifactorial causes, including physician's confidence in visual assessment from coronary angiograms,³ the need for inducing hyperemia, and the use of a costly pressure wire. Coronary angiography-based computational FFR methods have been developed as alternatives, based on combined geometrical data derived from coronary angiography and hemodynamic boundary conditions.^{4–7} Among them, quantitative flow ratio (QFR) is a novel approach for fast computation of FFR based on two angiographic views.^{4,8} QFR has been validated extensively showing superior diagnostic metrics^{9,10} and prognostic value.^{11,12}

The computation of QFR requires two angiographic views acquired at least 25° in separation with minimal vessel foreshortening and overlap, which is not always possible due to the curved course of coronary arteries. In addition, current QFR solutions assume a linear tapering of the reference vessel size, which might decrease the diagnostic accuracy in assessment of bifurcation lesions. Finally, automation of the lumen contour delineation and frame counting might further simplify QFR analysis and hence, increase its availability by default for routine use in the catheterization laboratory.

In this study, we developed a new method for computing QFR from a single angiographic view, empowered by artificial intelligence (AI) for automatic delineation of major coronary arteries and their side branches. A more accurate reconstruction of reference vessel size was achieved by calculating a step-down reference diameter function based on the Murray bifurcation fractal law. The diagnostic performance of this new Murray law-based QFR (μ QFR) was evaluated using wire-based FFR as the reference standard.

2 | METHODS

2.1 | Study design and patient population

This study performed a post-hoc analysis of all patients enrolled in the FAVOR (Functional Diagnostic Accuracy of Quantitative Flow Ratio in Online Assessment of Coronary Stenosis) II China study (NCT03191708). The FAVOR II China is a prospective, multicenter trial designed to evaluate the diagnostic accuracy of QFR in identifying hemodynamically significant coronary artery disease using wire-based

FFR as the reference standard.⁹ Adults with suspected or known coronary artery disease, who were admitted for diagnostic coronary angiography due to high risk of significant coronary stenosis, were consecutively enrolled. FFR was assessed in vessels with percent diameter stenosis between 30% and 90% in a vessel ≥ 2 mm by visual estimation. Detailed study design, endpoints, and inclusion and exclusion criteria have been published elsewhere.⁹

The study protocol complied with the Declaration of Helsinki and Good Clinical Practice guidelines of the China Food and Drug Administration. The study protocol was approved by the institutional review board and all patients provided written informed consent.

2.2 | Coronary angiography and FFR measurement

Details of the acquisition of angiographic images and FFR measurement have been published elsewhere.⁹ In short, angiographic images were recorded at 15 frames/s by flat-panel angiographic systems (AXIOM Artis, Siemens Healthcare, Erlangen, Germany; Innova, GE, Wauwatosa, Wisconsin; AlluraXper, Philips, Amsterdam, the Netherlands; INTEGRIS Allura, Philips). A table with recommended angiographic views was provided to the operators prior to angiographic image acquisition. Injection of the contrast medium was performed manually with a forceful and stable injection or by the pump at a rate of approximately 4 ml/s.

All FFR measurements were performed by using RadiAnalyzer Xpress instrument and Certus pressure wire (St. Jude Medical, St. Paul, Minnesota). Hyperemia was induced by intravenous administration of adenosine-5'-triphosphate (ATP) at ≥ 160 μ g/l/min. Pressure data were recorded for at least 3 s of stable value before ATP administration and for at least 10 s of stable value during hyperemia. The pressure sensor was returned to the guiding catheter tip to exclude pressure drift. A limited pressure drift within 0.05 during hyperemia was accepted; otherwise, the procedure was repeated. Notably, for FFR values between 0.75 and 0.85, only a smaller drift of 0.02 was considered acceptable.

2.3 | Analysis of μ QFR and quantitative coronary angiography

Computation of μ QFR was performed using a prototype software (AngioPlus Core, Pulse Medical Imaging Technology, Shanghai, China)

by an experienced analyst who was blinded to FFR data. The computation of μ QFR can be summarized with a few steps: (a) delineation of the interrogated epicardial coronary artery during contrast injection and calculation of contrast flow velocity based on the centerline length divided by the contrast dye-filing time;¹³ (b) selection of the analysis frame with sharp lumen contour at the stenotic segment as key frame; (c) delineation of the lumen contour of the interrogated vessel and its side branches with diameters of ≥ 1.0 mm on the key frame; (d) reconstruction of reference diameter function with step-down size across bifurcations according to Murray fractal law;^{14,15} (e) modeling of hyperemic flow velocity based on the contrast flow velocity and calculation of pressure drop based on fluid dynamics Equations,¹⁶ assuming a blood density of $1,060 \text{ kg/m}^3$ and viscosity of $0.0035 \text{ kg/(m}\cdot\text{s)}$. Specifically, the pressure loss caused by the frictional loss along the lesion entrance and stenotic segment as well as the inertial loss stemming from the sudden expansion of the flow as it emerged from the stenosis were calculated, based on the stenosis geometry and the hyperemic flow rate. In the end μ QFR was available for both the interrogated vessel and its side branches. During the same calculation, quantitative coronary angiography (QCA) data were also available from the software. Of note, delineation of lumen contours on the interrogated vessels and their side branches was performed automatically by AI using a convolutional neural network based on the U-Net architecture.¹⁷

From the two angiographic views of each interrogated vessel that were initially selected for QFR analysis in the FAVOR II China study, the analyst chose the angiographic view with minimal vessel overlap on both the interrogated vessel and its side branches ostia as the optimal angiographic view. The other angiographic view was then defined as the suboptimal angiographic view. Analysis of μ QFR and QCA was performed separately on these two angiographic views.

2.4 | Intra- and inter-observer analysis

To analyze intra-observer and inter-observer variability in μ QFR analysis, 50 patients were randomly selected and reanalyzed by the same analyst 1 week later and by a second analyst, following the same standard operation procedure and being blinded to each other or to the previous computational results.

2.5 | Statistical analysis

Continuous variables were tested for normal distribution by Shapiro-Wilk test and were reported as mean $\pm$ standard deviation if normally distributed or as median [interquartile range (IQR)] if non-normally distributed. Categorical variables were reported as counts (percentage). Spearman's correlation coefficient was used for correlation analysis. Bland-Altman analysis and intraclass correlation coefficient for the absolute value (ICCa) were used to test the agreement between different continuous variables. The between-center heterogeneity for assessment of the mean agreement between μ QFR and FFR was tested by I^2 statistics.

The area under the curve (AUC) by receiver-operating characteristic (ROC) curve analysis by Delong method was used to compare the accuracy of μ QFR and percent diameter stenosis (DS%) in predicting $\text{FFR} \leq 0.80$. The Youden index was used as the criterion to determine the best cut-off value for DS% in predicting $\text{FFR} \leq 0.80$. All statistical analyses were performed with MedCalc (version 14.12, MedCalc Software, Ostend, Belgium). A 2-sided value of $p < .05$ was considered as statistically significant.

3 | RESULTS

3.1 | Baseline clinical and lesion characteristics

A total of 330 vessels with both angiography and FFR data from 306 patients enrolled from 5 hospitals in the FAVOR II China study were analyzed in the core laboratory. Detailed study flowchart of FAVOR II China study, and baseline clinical and lesion characteristics have been published elsewhere.⁹ μ QFR was successfully computed in all 330 vessels of 306 patients. FFR had a mean value of 0.82 ± 0.12 and a median value of 0.85 [IQR: 0.77 to 0.90]. $\text{FFR} \leq 0.80$ was identified in 120 (36.4%) vessels by analyzing the pressure tracings in the core laboratory.

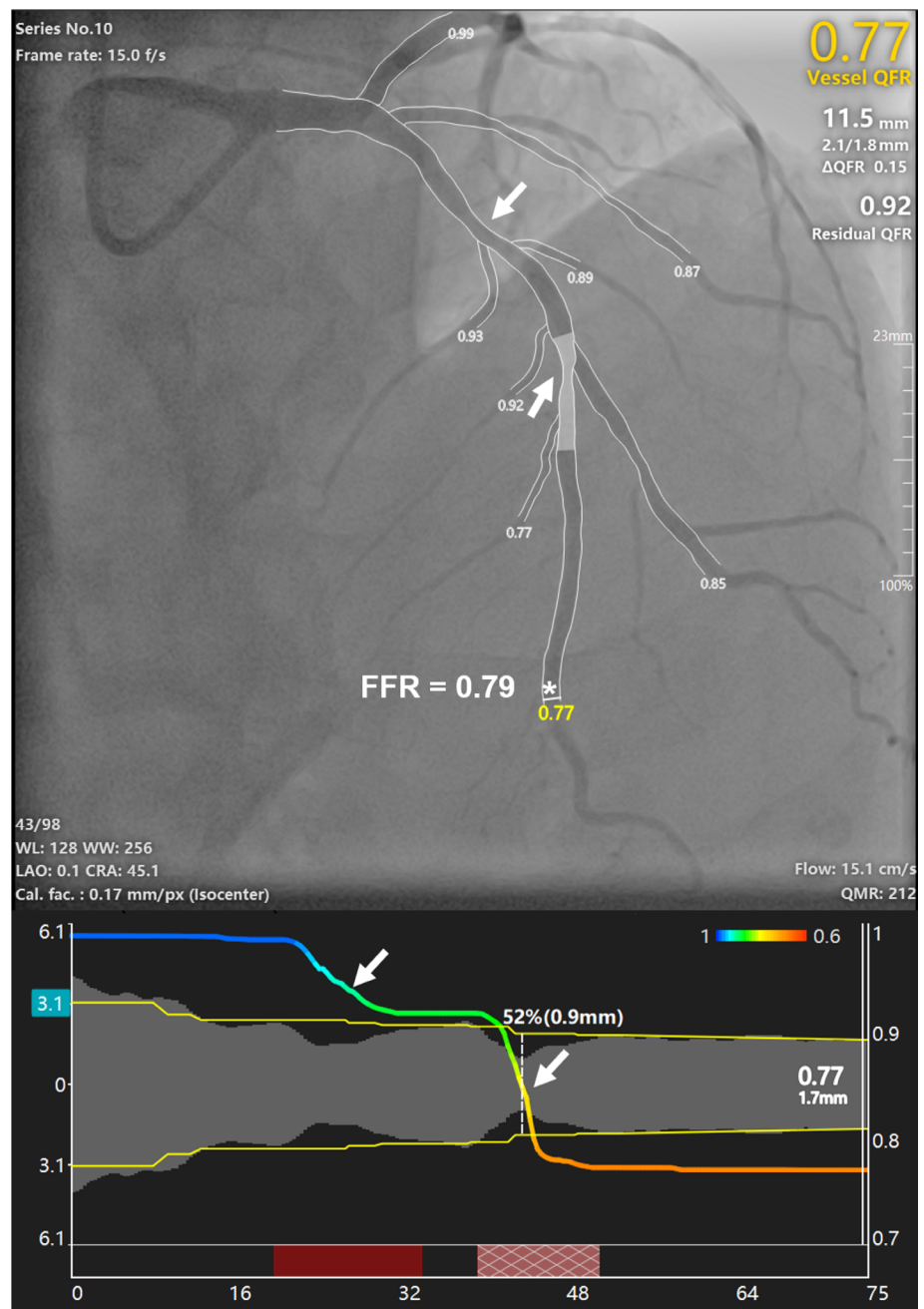
3.2 | Correlation and agreement between μ QFR and FFR

A representative example of computation of μ QFR is shown in Figure 1. μ QFR had a mean value of 0.82 ± 0.12 and a median value of 0.85 [IQR: 0.76 to 0.91]. μ QFR ≤ 0.80 was identified in 113 (34.2%) vessels. μ QFR showed excellent correlation ($r = 0.90$, $p < .001$) and agreement (mean difference = 0.00 ± 0.05 , $p = 0.378$; ICCa = 0.91 [95% CI: 0.89 to 0.92]) with FFR (Figure 2). The I^2 statistic for assessment of the mean agreement between μ QFR and FFR was 0.00 ($p = .997$), indicating that the between-center variance component was small enough to be ignored.

3.3 | Diagnostic performance of μ QFR

Using a cut-off value of ≤ 0.80 for FFR to define hemodynamically significant lesions, the vessel-level diagnostic accuracy of μ QFR was 93.0% (95% CI: 90.3 to 95.8%), with 105 true positives, 202 true negatives, 8 false positives, and 15 false negatives. The sensitivity, specificity, positive predictive value, negative predictive value, positive likelihood ratio, and negative likelihood ratio for μ QFR ≤ 0.80 to identify $\text{FFR} \leq 0.80$ was 87.5% (95% CI: 80.2 to 92.8%), 96.2% (95% CI: 92.6 to 98.3%), 92.9% (95% CI: 86.5 to 96.9%), 93.1% (95% CI: 88.9 to 96.1%), 23.0 (95% CI: 11.6 to 45.5), and 0.13 (95% CI: 0.08 to 0.20), respectively (Table 1). The AUC in predicting hemodynamically significant stenosis was significantly higher for μ QFR ≤ 0.80 than QCA-derived DS% $\geq 50\%$ (0.97 [95% CI: 0.95 to 0.99], $p < .001$ versus

FIGURE 1 A representative example of computation of μ QFR. Upper panel shows two sequential intermediate lesions at the LAD. The lumen contours of LAD and its side branches were automatically delineated and superimposed on the angiographic images. The computed μ QFR at the asterisk position was 0.77 while the wire-based FFR was 0.79. Lower panel shows the co-registration between lumen size and μ QFR pullback at every position along the LAD. The two intermediate lesions correspond to two significant pressure drops in the μ QFR pullbacks (white arrows). FFR, fractional flow reserve; LAD, left anterior descending artery; QFR, quantitative flow ratio; μ QFR, Murray law-based QFR



0.83 [95% CI: 0.78 to 0.87], $p < .001$, difference: 0.14 [95% CI: 0.10 to 0.19], $p < .001$) (Figure 3). The best cut-off value for DS% in determining $\text{FFR} \leq 0.80$ defined by Youden index was $\geq 48\%$.

3.4 | Diagnostic performance of μ QFR using suboptimal angiographic image projection

Use of suboptimal angiographic image projection decreased the diagnostic accuracy of μ QFR (AUC = 0.97 versus 0.92, difference = 0.05, $p < .001$) (Figure 3). Table 2 summarizes the diagnostic performance of μ QFR using suboptimal angiographic image projection.

3.5 | Reproducibility in repeated μ QFR analysis

Repeated μ QFR analysis was performed in 50 vessels from 49 patients. Intra-observer and inter-observer variability in μ QFR was 0.00 ± 0.03 and 0.00 ± 0.03 , respectively.

3.6 | Computational performance of μ QFR

Average analysis time from the moment when the coronary angiographic view was loaded in the software until μ QFR computation was completed was 67 ± 22 s on an off-the-shelf computer with Intel

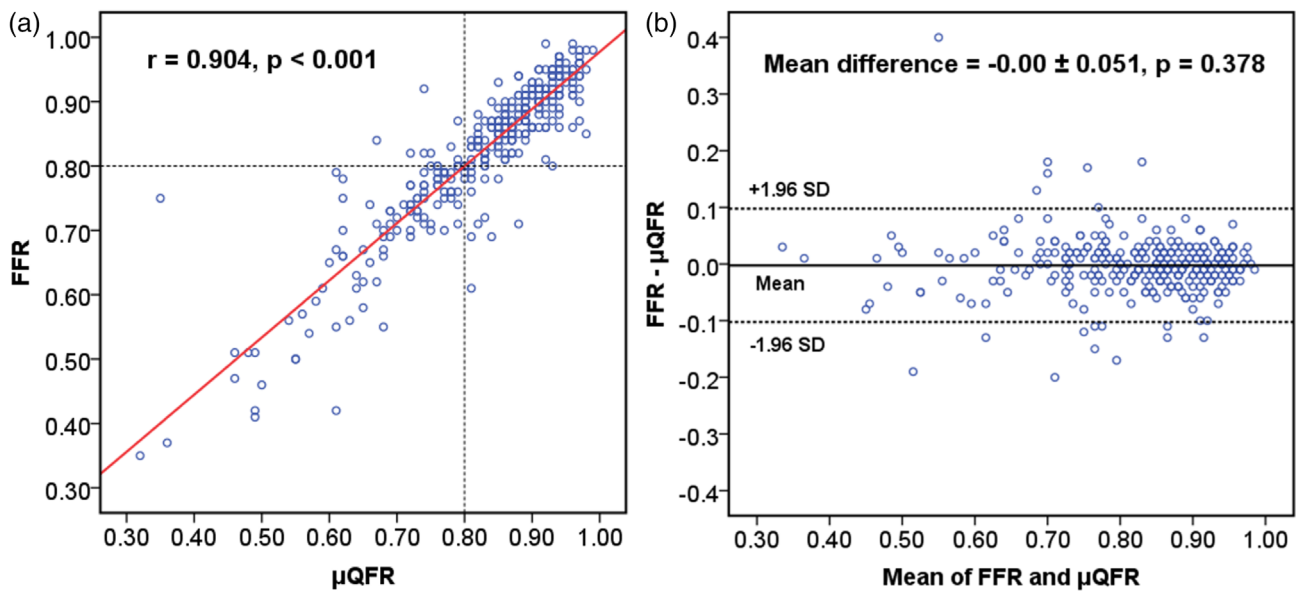


FIGURE 2 Correlation and agreement between μ QFR and FFR. Excellent correlation (left panel) and agreement (right panel) was observed between μ QFR and FFR

TABLE 1 Diagnostic performance of μ QFR and QCA-derived DS% in predicting FFR ≤ 0.80

	μ QFR ≤ 0.80	DS% $\geq 50\%$
Accuracy, % (95% CI)	93.0 (90.2, 95.8)	76.1 (71.4, 80.7)
Sensitivity, % (95% CI)	87.5 (80.2, 92.8)	57.5 (48.1, 66.5)
Specificity, % (95% CI)	96.2 (92.6, 98.3)	86.7 (81.3, 91.0)
PPV, % (95% CI)	92.9 (86.5, 96.9)	71.1 (61.0, 79.9)
NPV, % (95% CI)	93.1 (88.9, 96.1)	78.1 (72.2, 83.2)
+LR (95% CI)	23.0 (11.6, 45.5)	4.3 (3.0, 6.3)
-LR (95% CI)	0.13 (0.08, 0.20)	0.49 (0.40, 0.60)

Abbreviations: Data for +LR and -LR are presented as ratio (95% CI) and for the rest of parameters as % (95% CI). CI, confidence interval; DS%, percent diameter stenosis; FFR, fractional flow reserve; +LR, positive likelihood ratio; -LR, negative likelihood ratio; NPV, negative predictive value; PPV, positive predictive value; μ QFR, Murray law-based quantitative flow ratio; QCA, quantitative coronary angiography.

i7-9750H processor (Intel Corporation, Santa Clara, CA; 2.6 GHz), 32 GB of RAM and 6G of GPU memory (NVIDIA GeForce RTX 2060).

4 | DISCUSSION

We have developed a new method for computation of FFR from a single angiographic view, empowered by AI in automatic delineation of lumen contour of major epicardial coronary arteries and their side branches. The Murray bifurcation fractal law was used to reconstruct more accurate reference vessel size for the computation of μ QFR. The main findings of the present study can be summarized as follows: (a) μ QFR showed excellent correlation ($r = 0.90$, $p < .001$) and agreement (mean difference = 0.00 ± 0.05 , $p = .378$) with wire-based FFR;

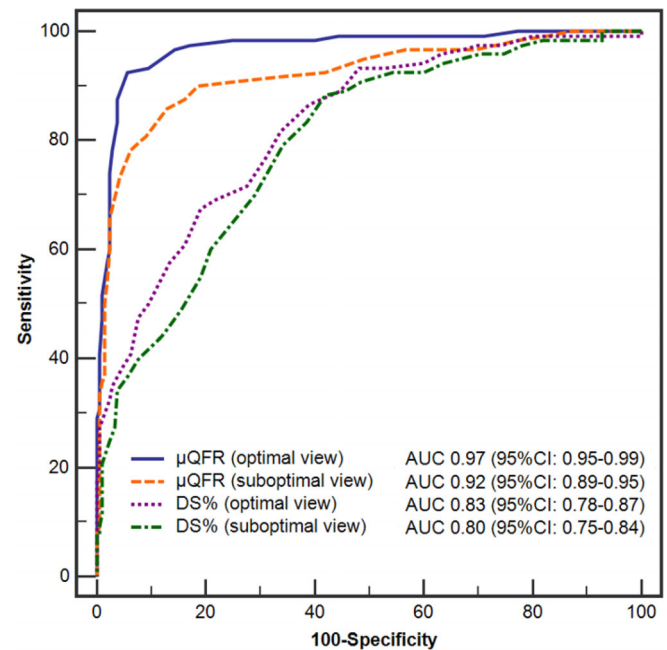


FIGURE 3 Comparison of diagnostic performance between μ QFR and QCA-derived DS% calculated using optimal and suboptimal angiographic view by ROC analysis. The μ QFR computed from the optimal angiographic view had the highest AUC in predicting FFR ≤ 0.80 , followed by μ QFR computed from the suboptimal angiographic view, DS% from the optimal angiographic view, and DS% from the suboptimal angiographic view. AUC, area under the ROC curve; ROC, receiver-operating characteristics; DS%, percent diameter stenosis; other abbreviations as in Figure 1

(b) the vessel-level diagnostic accuracy for μ QFR ≤ 0.80 in predicting FFR ≤ 0.80 was 93.0%, with sensitivity of 87.5% and specificity of 96.2%; (c) the use of suboptimal angiographic image view decreased

TABLE 2 Diagnostic performance of μ QFR and QCA-derived DS% using suboptimal angiographic image projection in predicting FFR ≤ 0.80

	μ QFR ≤ 0.80	DS% $\geq 50\%$
Accuracy, % (95% CI)	88.2 (84.7, 91.7)	71.8 (66.9, 76.7)
Sensitivity, % (95% CI)	78.3 (69.9, 85.3)	46.7 (37.5, 56.0)
Specificity, % (95% CI)	93.8 (89.6, 96.7)	86.2 (80.8, 90.6)
PPV, % (95% CI)	87.9 (80.1, 93.4)	65.9 (54.8, 75.8)
NPV, % (95% CI)	88.3 (83.4, 92.2)	73.9 (67.9, 79.3)
+LR (95% CI)	12.7 (7.4, 21.6)	3.4 (2.3, 5.0)
-LR (95% CI)	0.23 (0.20, 0.30)	0.62 (0.50, 0.70)

Abbreviations: Data for +LR and -LR are presented as ratio (95% CI) and for the rest of parameters as % (95% CI). CI, confidence interval; DS%, percent diameter stenosis; FFR, fractional flow reserve; +LR, positive likelihood ratio; -LR, negative likelihood ratio; NPV, negative predictive value; PPV, positive predictive value; μ QFR, Murray law-based quantitative flow ratio; QCA, quantitative coronary angiography.

the diagnostic accuracy of μ QFR (AUC = 0.97 versus 0.92, $p < .001$); (d) low intra- and inter-observer variability was observed in repeated μ QFR analysis; (e) analysis of μ QFR had an average analysis time of 67 ± 22 s.

The FAVOR II China study showed that QFR computation based on the 3-dimensional (3D) reconstruction of two angiographic views had a vessel-level diagnostic accuracy of 92.7% (95% CI: 89.3 to 95.3%) with online in-procedure analysis and 93.3% (95% CI: 90.0 to 95.7%) with offline core laboratory analysis.⁹ The present study found that μ QFR had a comparable diagnostic accuracy by computation of QFR based on a single angiographic view without the need to perform 3D angiographic reconstruction. Several reasons might explain this finding. Firstly, the lumen contour of the side branches was delineated in μ QFR computation and, instead of assuming a linear tapering of the reference vessel size, a step-down reference diameter function of the hypothesized healthy vessel was reconstructed. This captured the natural bifurcation fractal physiology and contributed to more accurate quantification of the extent of lesion severity, which is the key factor to determine FFR. Similar finding was observed in the recent validation of optical flow ratio, that is, the FFR computation from optical coherence tomography.¹⁹ Secondly, the angiographic view with optimal image quality was used for μ QFR computation. Optimal exposure of the stenotic segment facilitates the quantification of the interrogated vessel geometry. Theoretically, 3D angiographic reconstruction based on two angiographic views can be more accurate in the quantification of lumen area when the lesions were eccentric. However, a second angiographic view might also introduce inaccurate sizing if there was vessel foreshortening and overlap in the second angiographic view. Indeed, the present study showed that the diagnostic performance of μ QFR based on a suboptimal angiographic view was slightly impaired. This emphasizes the importance of using angiographic views with optimal exposure of all lesions in the interrogated vessel for the computation of μ QFR. Finally, the quantification of contrast flow velocity in the computation of μ QFR based on a single angiographic view might

indeed be influenced by vessel foreshortening in individual patients. Nevertheless, previous studies demonstrated that such variation in the contrast flow velocity had negligible impact on the diagnostic accuracy of QFR.¹³

Notably, the reconstruction of the step-down reference vessel size also improved the diagnostic concordance between QCA-derived DS% and FFR. In the FAVOR II China study, DS% by conventional QCA-derived DS% with linear tapering of the reference diameter function had an accuracy of only 64.0% in predicting FFR ≤ 0.80 .⁹ By using step-down reference diameter function, the present study found that the diagnostic accuracy of QCA-derived DS% improved to 76.1%. Even with a suboptimal angiographic view, the diagnostic accuracy of QCA-derived DS% reached 71.8%. This demonstrated the necessity of using the step-down reference vessel size in the quantification of coronary stenosis severity.²⁰ The accurate reference vessel dimension would also enable more accurate 2-dimensional QCA analysis for bifurcation lesions.

4.1 | Feasibility and efficiency in μ QFR computation

Despite that the use of FFR for identifying flow limiting coronary stenosis has been recommended by major guidelines,¹ the clinical adaption of FFR remains low in many countries.² Several underlying reasons might explain this gap. In addition to the cost of disposable pressure wire and the discomfort caused by hyperemia, it was reported that a major obstacle to the adoption of coronary physiological assessment in the catheterization laboratory lies in physician's confidence in visual assessment of angiographic images alone.³ Thus, image-based computational FFR could potentially contribute to wider adoption of coronary physiological assessment.²¹ However, the feasibility and efficiency of computational FFR is crucial for it to be widely adopted in routine use in the catheterization laboratory. In previous validation studies, failure to acquire two satisfactory angiographic views would impair the feasibility of QFR analysis.^{9,10,22} Operators were encouraged to acquire angiographic images according to recommended angiographic projections, thus increasing the feasibility of QFR in prospective validation studies.^{9,22} Nevertheless, in the prospective WIFI II study, online in-procedure QFR analysis was still unsuccessful in 5% of the total vessels mainly due to presence of severe vessel overlap or foreshortening in at least one of the angiographic views.¹⁰ The same limitation was found in studies investigating the feasibility of QFR computation in post-PCI settings.¹¹ Computation of μ QFR based on a single angiographic view can resolve such a limitation, especially in patients with tortuous, sequential, or diffuse lesions where it was not always possible to obtain two angiographic views both with good exposure of all lesions in the interrogated vessels. Furthermore, μ QFR was available in both the main vessel and all major side branches. Thus, the analysis time in patients with multi-vessel disease can be further reduced compared with the existing QFR method, which can only analyze one vessel at the time. Finally, the use of AI in automation of the lumen contouring and frame counting decreased the need for manual correction in the QFR analysis. Average

analysis time of μ QFR was about 1 min in the present study, which was significantly faster than the existing QFR computation with reported median analysis time of 5 min in the FAVOR II Europe-Japan study.²² This might be the key step to reduce the training time of QFR analysis while maintaining good reproducibility.

At present, utility of FFR-guided PCI of bifurcation lesions is still limited to selected centers. Studies demonstrated the superiority of FFR over coronary angiography in guiding PCI of bifurcation lesions by reducing unnecessary complicated side branch intervention and the incidence of main vessel restenosis.^{23,24} However, it is technically challenging to perform FFR measurement of a jailed side branch after stenting the main vessel. Previous studies reported 5% to 10% rate of failure when physicians tried to pass the pressure wire through stent cells to the side branch.²⁴ μ QFR can be a useful alternative in these scenarios, providing simultaneous anatomical and physiological assessment for all the side branches while exempting from the use of pressure wire. With μ QFR, the physiological significance of the side branches can be obtained simultaneously with the analysis of the main vessel. This substantially increases the feasibility of physiology-guided bifurcation PCI. Furthermore, since the computation of μ QFR used a step-down reference vessel, it can provide more accurate assessment of lesion when it involves both the proximal main vessel and the side branches, which is less accurate in conventional single-vessel QFR analysis and was excluded in the previous QFR validation studies.

LIMITATIONS

This study is not without limitations. The analysis of μ QFR was performed offline at a core laboratory, the diagnostic accuracy of μ QFR when performed by medical staff during the procedure in the catheterization laboratory still needs to be assessed. In addition, although μ QFR can be used to assess the physiological significance of all the side branches, wire-based FFR was not measured in the FAVOR II China study. Therefore, further studies to validate the diagnostic accuracy of μ QFR in side branches are warranted. Finally, we were not able to perform comparison between μ QFR computed from a single angiographic view versus that from two angiographic views since the prototype software did not support reconstruction of all side branches in 3D.

CONCLUSIONS

Computation of μ QFR from a single angiographic view has high feasibility and excellent diagnostic accuracy in identifying hemodynamically significant coronary stenosis. The short analysis time and good reproducibility of μ QFR bear the potential of wider adoption of physiological assessment in the catheterization laboratory.

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CONFLICT OF INTEREST

Yunxiao Chang is an employer of Pulse Medical Imaging Technology. Shengxian Tu received research grants from Pulse Medical Imaging Technology. Other authors report no conflicts of interest regarding this manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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